

Module 2: Transport Layer and Network Layer



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Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
 - select()
 - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

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- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using ip

4.2 Network-Layer Protocols

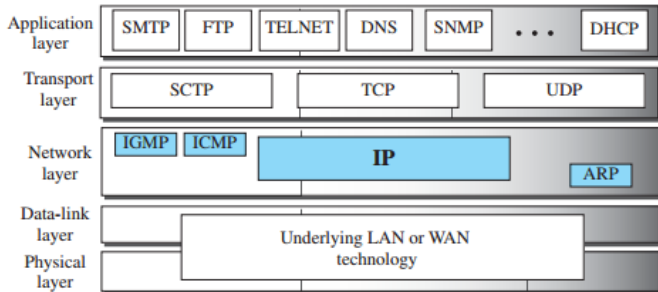
Network Layer in TCP/IP

- Implements communication between source and destination across interconnected networks.
- Current Internet primarily uses **IPv4**; **IPv6** is the next-generation protocol.

Protocols at the Network Layer

Protocol	Function
IPv4	Packetizing, forwarding and delivery of datagrams
ICMPv4	Reports errors and provides diagnostic messages
IGMP	Supports multicast communication
ARP	Maps IPv4 addresses to Data-Link (MAC) addresses

Figure 4.23 *Position of IP and other network-layer protocols in TCP/IP protocol suite*



IPv4 is a Connectionless Datagram Protocol

- No connection is established before transmission.
- Every datagram is treated independently.
- Different packets of the same message may:
 - Follow different routes.
 - Arrive out of order.
 - Experience different delays.
- Routing decisions are made independently for each packet.
- Ordering and reliability are handled by higher-layer protocols.

Best-Effort Delivery Service

IPv4 provides Best-Effort Delivery

- No guarantee that packets will reach the destination.
- Packets may be:
 - Lost
 - Delayed
 - Corrupted
 - Delivered out of order
- IPv4 does not perform recovery or retransmission.
- Reliable delivery is provided by transport-layer protocols such as **TCP**.

Topics Covered in This Chapter

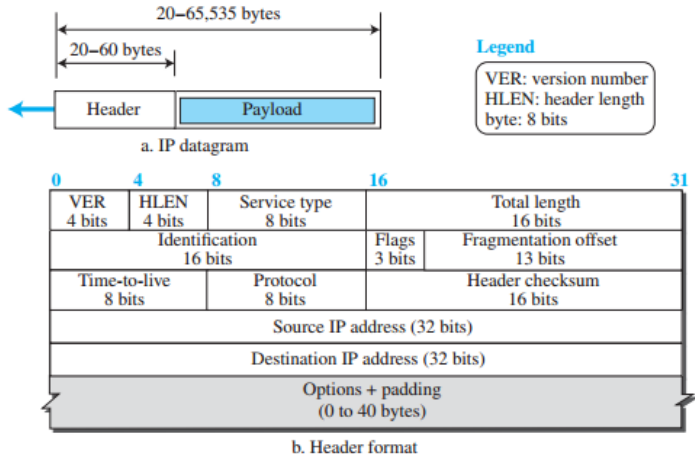
- 1 IPv4 Datagram Format
- 2 IPv4 Addressing
- 3 Routing and Forwarding
- 4 ICMP

IPv4 Datagram Format

IPv4 Datagram

- IP packets are called **Datagrams**.
- A datagram is a **variable-length packet**.
- Consists of:
 - **Header** (20–60 bytes)
 - **Payload (Data)**
- Header contains routing, forwarding and delivery information.
- Header is represented in **4-byte words**.

Figure 4.24 *IP datagram*



IPv4 Header Fields (1/3)

Version (4 bits)

- Specifies IP version.
- IPv4 value = 4.

Header Length (HLEN)

- Size of header in 4-byte words.
- Actual Header Length = $\text{HLEN} \times 4$ bytes.
- Indicates where payload begins.

Service Type

- Earlier called TOS.
- Now used for **DiffServ (QoS)**.

Total Length

- Header + Payload.
- Maximum = **65,535 bytes**.

Identification

- Identifies fragmented datagrams.

Flags

- Control fragmentation.
- DF (Don't Fragment)
- MF (More Fragments)

Fragment Offset

- Position of fragment in original datagram.

IPv4 Header Fields (2/3)

Time To Live (TTL)

- Prevents packets from looping forever.
- Decrement by every router.
- Packet discarded when TTL becomes zero.

Protocol

- Identifies upper-layer protocol.
- Examples:
 - TCP
 - UDP
 - ICMP
- Supports multiplexing/demultiplexing.

Header Checksum

- Detects header errors only.
- Payload checked by TCP/UDP.
- Recomputed at every router.

Source Address

- 32-bit sender IP address.

Destination Address

- 32-bit receiver IP address.
- Remains unchanged during transmission.

IPv4 Header Fields (3/3)

Options

- Optional field (0–40 bytes).
- Used for:
 - Network testing
 - Debugging
 - Diagnostics
- Routers may modify options.
- Header checksum must be recalculated.

Payload

- Actual user data.
- Carries packets from:
 - TCP
 - UDP
 - ICMP
 - Routing protocols
- Similar to the contents of a postal package.

Key Point

IPv4 checks only the **header**. Error detection of payload is handled by upper-layer protocols.

IPv4 Datagram Header Summary

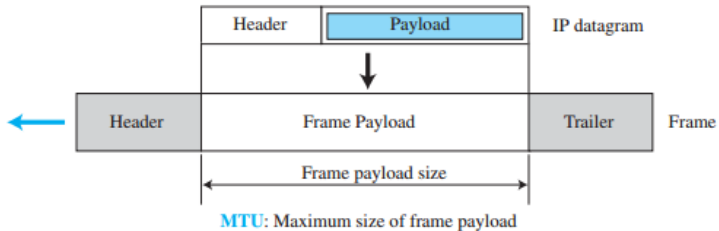
Field	Purpose
Version	Specifies IPv4 version
HLEN	Header length
Service Type	QoS / DiffServ
Total Length	Total datagram size
Identification	Datagram identification
Flags	Fragmentation control
Fragment Offset	Position of fragment
TTL	Limits packet lifetime
Protocol	Upper-layer protocol
Header Checksum	Header error detection
Source Address	Sender IP address
Destination Address	Receiver IP address
Options	Testing and debugging
Payload	User data

IPv4 Fragmentation

What is Fragmentation?

- An IP datagram may travel through different networks having different frame sizes.
 - Every router removes the incoming frame, processes the IP datagram, and encapsulates it into a new outgoing frame.
 - If the outgoing network cannot carry the entire datagram, it is divided into smaller fragments.
 - This process is called **Fragmentation**.
-
- Fragmentation may occur:
 - At the source host
 - At any router along the path
 - Reassembly is performed **only at the destination host**.

Figure 4.26 *Maximum transfer unit (MTU)*



Maximum Transfer Unit (MTU)

Maximum Transfer Unit (MTU)

- Every link-layer protocol defines a maximum payload size called the **MTU**.
- IP datagram size must be less than or equal to the MTU.
- MTU varies for different networks.
- Typical LAN MTU $\equiv$ **1500 bytes**
- IPv4 supports datagrams up to **65,535 bytes**.
- Large datagrams are fragmented before transmission.
- A fragment can itself be fragmented if it encounters a smaller MTU.

Key Point

Fragmentation allows IPv4 to work across heterogeneous networks having different MTU values.

Characteristics of Fragmentation

- Each fragment is treated as an independent IP datagram.
- Every fragment contains:
 - Its own IP header
 - A portion of the original payload
- Most header fields are copied to every fragment.
- The following fields are modified:
 - Flags
 - Fragment Offset
 - Total Length
 - Header Checksum (recalculated)
- Fragments may travel through different routes.
- Destination waits for all fragments before reassembly.

Fragmentation Fields

Identification (16 bits)

- Uniquely identifies each original datagram.
- Same value copied into every fragment.
- Helps destination reassemble fragments.

Flags (3 bits)

- First bit: Reserved.
- DF (Don't Fragment).
- MF (More Fragments).

DF Bit

- DF=1 : Fragmentation prohibited.
- Router discards packet if fragmentation is required.
- ICMP error sent to sender.

MF Bit

- MF=1 : More fragments follow.
- MF=0 : Last (or only) fragment.

Fragment Offset

Fragment Offset

- Indicates the position of a fragment within the original datagram.
- Offset is measured in units of **8 bytes**.
- Allows destination to place fragments in correct order.

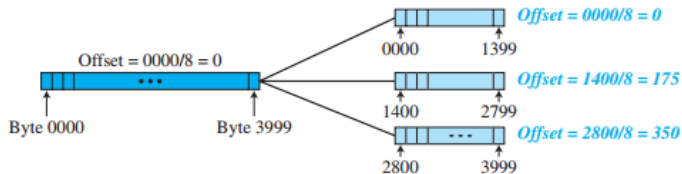
Example

Fragment	Data Bytes	Offset
1	0 – 1399	0
2	1400 – 2799	175
3	2800 – 3999	350

Important

Offset = Starting Byte Number / 8

Figure 4.27 *Fragmentation example*



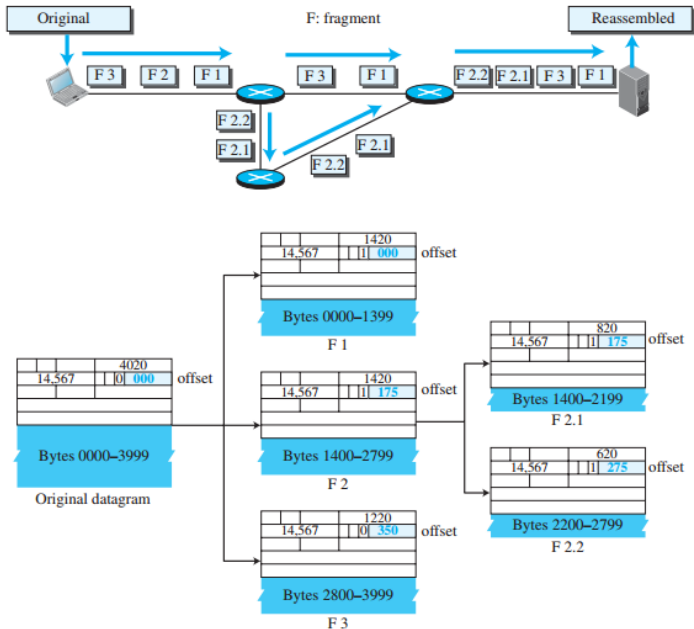
Fragment Reassembly

- Reassembly is performed only at the destination host.
- All fragments are identified using:
 - Identification field
 - Fragment Offset
 - MF flag
- Fragments may arrive:
 - Out of order
 - Through different paths
- Destination correctly reconstructs the original datagram.

Reassembly Steps

- 1 First fragment has Offset = 0.
- 2 Next fragment offset = Previous Fragment Length / 8.
- 3 Continue until fragment with MF = 0 is received.

Figure 4.28 Detailed fragmentation example



Fragmentation Summary

Concept	Description
MTU	Maximum payload supported by a network
Fragmentation	Splitting a large datagram into smaller fragments
Fragmentation Location	Source host or intermediate router
Reassembly	Destination host only
Identification	Common ID for all fragments
DF Flag	Prevents fragmentation
MF Flag	Indicates more fragments follow
Fragment Offset	Position of fragment (8-byte units)
Checksum	Recalculated for every fragment

Exam Point

Every fragment is an independent IP datagram with its own header, but all fragments carry the same Identification value for successful reassembly.

Security Issues in IPv4

Why is IPv4 Security Needed?

- IPv4 was originally designed for a trusted Internet environment.
- No built-in security mechanisms were included.
- Modern Internet communication faces several security threats.

Major Security Threats

- 1 Packet Sniffing
- 2 Packet Modification
- 3 IP Spoofing

Need for Security

IPv4 alone cannot protect against these attacks. Additional security mechanisms such as **IPSec** are required.

Common IPv4 Security Attacks

1. Packet Sniffing

- Passive attack.
- Attacker copies intercepted packets.
- Packet contents are not modified.
- Difficult to detect.
- Prevented using **Encryption**.

2. Packet Modification

- Attacker alters packet contents.
- Receiver assumes packet is genuine.
- Detected using **Data Integrity** checks.

3. IP Spoofing

- Attacker forges the source IP address.
- Pretends to be another legitimate host.
- Example:
 - Fake bank customer request.
- Prevented using **Origin Authentication**.

IPSec

- Security protocol used along with IPv4.
- Creates a secure communication channel between two entities.
- Protects against:
 - Packet Sniffing
 - Packet Modification
 - IP Spoofing

Services Provided by IPSec

① Algorithm and Key Management

- Selects encryption algorithms and shared keys.

② Packet Encryption

- Protects data confidentiality.
- Prevents packet sniffing.

③ Data Integrity

- Detects unauthorized packet modifications.

④ Origin Authentication

- Verifies sender identity.
- Prevents IP spoofing.

IPv4 Addresses

Definition

- An **IPv4 address** is a **32-bit logical address** used to identify the connection of a host or router to the Internet.
- It uniquely identifies a **network interface (connection)**, not the device itself.
- If a device is moved to another network, its IP address usually changes.

Characteristics

- **Unique:** Every Internet connection has a unique IPv4 address.
- **Universal:** The addressing scheme is accepted worldwide.
- A device with multiple network interfaces has multiple IP addresses.

Key Point

IPv4 identifies the **connection to the Internet**, not the physical computer.

Address Space and Address Notations

Address Space

- IPv4 uses **32 bits**.
- Total possible addresses:

$$2^{32} = 4,294,967,296$$

- More than **4 billion** unique addresses.

IPv4 Address Representations

- 1 Binary Notation (32 bits)
- 2 Dotted Decimal Notation
- 3 Hexadecimal Notation

Example

- Binary:

11000000 10101000 00000001 00001010

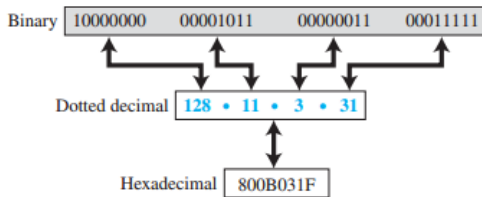
- Dotted Decimal:

192.168.1.10

- Hexadecimal:

C0A8010A

Figure 4.29 *Three different notations in IPv4 addressing*



Hierarchical Addressing

Why Hierarchical Addressing?

- Makes routing scalable.
- Similar to postal addresses and telephone numbers.

IPv4 Address Structure



- Prefix identifies the network.
- Suffix identifies the host (device connection).
- If prefix length is n ,

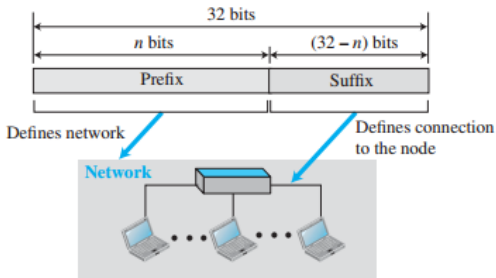
$$\text{Host bits} = 32 - n$$

Important

Prefix length may be:

- Fixed (Classful Addressing)
- Variable (Classless Addressing / CIDR)

Figure 4.30 *Hierarchy in addressing*



Classful Addressing

Original IPv4 Addressing Scheme

Class	First Bits	Prefix	Purpose
A	0	/8	Very Large Networks
B	10	/16	Medium Networks
C	110	/24	Small Networks
D	1110	–	Multicast
E	1111	–	Reserved

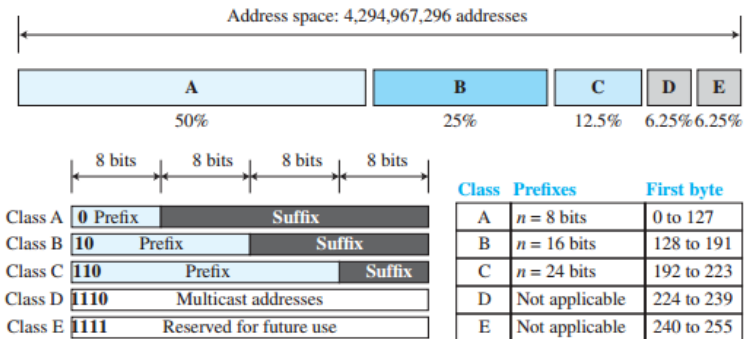
Number of Networks

- Class A: $2^7 = 128$ networks
- Class B: $2^{14} = 16,384$ networks
- Class C: $2^{21} = 2,097,152$ networks

Observation

Classes D and E are not divided into network and host portions.

Figure 4.31 Occupation of the address space in classful addressing



Problems with Classful Addressing

Address Depletion

- Classful addressing wasted a large number of IPv4 addresses.
- Address blocks were allocated in fixed sizes regardless of actual requirements.

Problems in Each Class

• Class A:

- Only 128 networks available.
- Each network supported over 16 million hosts.
- Most addresses remained unused.

• Class B:

- Suitable for medium organizations.
- Many allocated addresses remained unused.

• Class C:

- Only 256 addresses per network.
- Too small for many organizations.

- Class D and E addresses were mostly unused.

Subnetting, Supernetting and CIDR

Subnetting

- Divides one large network into multiple smaller subnetworks.
- Increases the prefix length.
- Example:

$$/8 \rightarrow /10$$

Supernetting

- Combines several Class C networks into one larger block.
- Intended for organizations needing more than 256 addresses.

Limitations

- Neither subnetting nor supernetting completely solved address exhaustion.

Solution

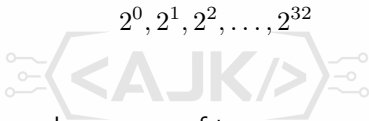
Internet authorities introduced **Classless Addressing (CIDR)** in 1996.

Classless Addressing (CIDR)

CIDR (Classless Inter-Domain Routing)

- Eliminates fixed address classes.
- Uses **variable-length prefixes**.
- Address blocks can contain:

addresses.

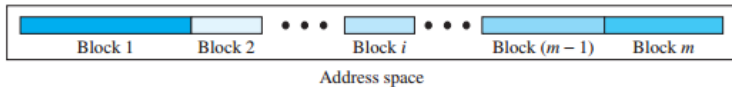


- Block size must always be a power of two.

Advantages

- Efficient address allocation.
- Reduced address wastage.
- Better routing scalability.
- Suitable for Internet Service Providers (ISPs).

Figure 4.32 *Variable-length blocks in classless addressing*



CIDR Slash Notation

Every CIDR address is written as

IP Address/ n

where

- n = Prefix length
- Network bits = n
- Host bits = $32 - n$

Examples

- 192.168.10.0/24
- 172.16.8.0/20
- 10.0.0.0/8

Observation

Unlike classful addressing, an IPv4 address alone does **not** identify its network. The prefix length must also be specified.

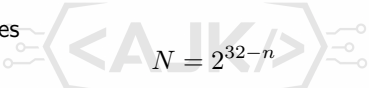
Extracting Information from a CIDR Address

Given:

Address/ n

we can determine:

- 1 Number of addresses


$$N = 2^{32-n}$$

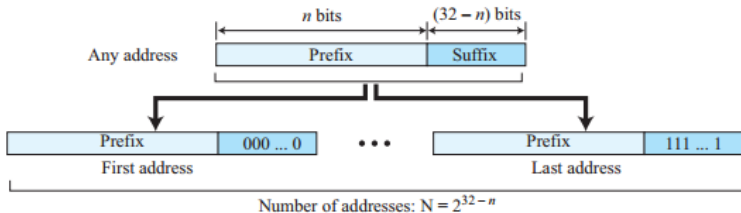
- 2 First address

Keep prefix bits and set all host bits to 0.

- 3 Last address

Keep prefix bits and set all host bits to 1.

Figure 4.34 *Information extraction in classless addressing*



Example 4.1: CIDR Block Calculation

Given

167.199.170.82/27

Step 1: Number of Addresses

$$N = 2^{32-27} = 2^5 = 32$$

Step 2: First Address

Keep first 27 bits and make remaining bits 0.

167.199.170.64/27

Step 3: Last Address

Keep first 27 bits and make remaining bits 1.

167.199.170.95/27

Result

- Block Size = 32 addresses
- First Address = 167.199.170.64
- Last Address = 167.199.170.95

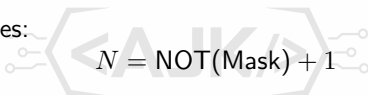
Address Mask

Address Mask

- A 32-bit number used to identify the network portion of an address.
- The first n bits are set to 1.
- The remaining $(32 - n)$ bits are set to 0.

Applications

- ① Number of addresses:


$$N = \text{NOT}(\text{Mask}) + 1$$

- ② First address:

Address AND Mask

- ③ Last address:

Address OR (NOT Mask)

Why use a Mask?

Bitwise AND, OR and NOT operations allow computers to efficiently determine network information.

Example 4.2: Using Address Mask

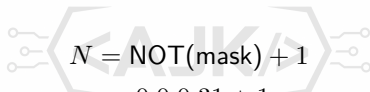
Given

167.199.170.82/27

Mask:

255.255.255.224

Calculations


$$\begin{aligned} N &= \text{NOT}(\text{mask}) + 1 \\ &= 0.0.0.31 + 1 \\ &= 32 \text{ addresses} \end{aligned}$$

First Address = Address AND Mask = 167.199.170.64

Last Address = Address OR (NOT Mask) = 167.199.170.95

Observation

Address masking provides the same result as directly using the CIDR prefix.

Example 4.3: One Address, Multiple Blocks

A classless address alone does not define its network.

Example:

230.8.24.56

Depending on the prefix length, it may belong to different blocks.

Prefix	Address Block
/16	230.8.0.0 – 230.8.255.255
/20	230.8.16.0 – 230.8.31.255
/26	230.8.24.0 – 230.8.24.63
/27	230.8.24.32 – 230.8.24.63
/29	230.8.24.56 – 230.8.24.63
/31	230.8.24.56 – 230.8.24.57

Important

An IPv4 address must always be accompanied by its prefix length.

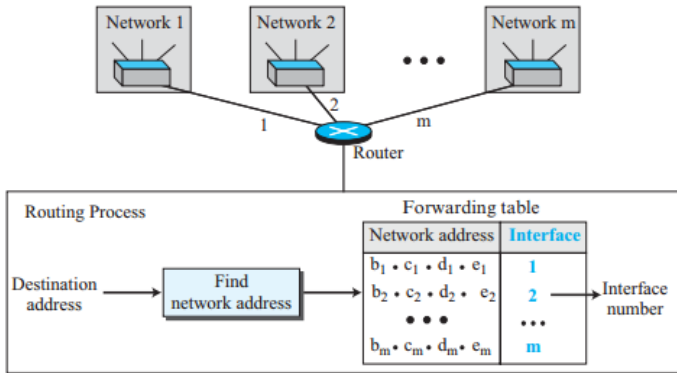
Network Address

- The first address of every CIDR block is called the network address.
- It uniquely identifies the network.
- Routers forward packets based on the network address.

Packet Forwarding

- 1 Extract destination network address.
- 2 Search forwarding table.
- 3 Determine output interface.
- 4 Forward packet toward destination network.

Figure 4.35 *Network address*



Block Allocation in CIDR

Who allocates addresses?

- ICANN allocates large address blocks.
- ISPs receive these blocks.
- ISPs allocate smaller blocks to organizations and customers.

CIDR Allocation Rules

- 1 Number of addresses must be a power of 2.
- 2 Addresses must be contiguous.
- 3 First address must be divisible by the block size.

Formula:

$$n = 32 - \log_2 N$$

$$\text{First Address} = (\text{Prefix}) \times 2^{32-n}$$

Example 4.4: Block Allocation

Problem

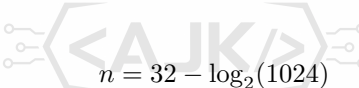
An ISP requests 1000 IPv4 addresses.

Solution

- 1000 is not a power of 2.
- Allocate the next power of 2:

$$1024 = 2^{10}$$

- Prefix length:


$$\begin{aligned}n &= 32 - \log_2(1024) \\ &= 32 - 10 = 22\end{aligned}$$

- Allocated block:

$18.14.12.0/22$

- First address is divisible by 1024, satisfying CIDR allocation rules.

Key Point

CIDR always allocates address blocks in powers of two.

Subnetting

Subnetting

- Subnetting divides a network into multiple smaller subnetworks (subnets).
- Each subnet has its own network address.
- Improves network organization and routing efficiency.
- An organization or ISP can create multiple hierarchy levels:
 - Network
 - Subnetwork
 - Sub-subnetwork
 - and so on.



Subnet Design Rules

- 1 Number of addresses must be a power of 2.
- 2 Prefix length:
$$n_{sub} = 32 - \log_2(N_{sub})$$
- 3 Starting address must be divisible by the subnet size.
- 4 Allocate larger subnetworks before smaller ones.

Example 4.5: Designing Subnets

Given Block

14.24.74.0/24

Total addresses:

$$2^{32-24} = 256$$

Required subnetworks:

- 120 hosts
- 60 hosts
- 10 hosts



Step 1: Allocate Largest Subnet First

Required	Allocated	Prefix
120	128	/25
60	64	/26
10	16	/28

Example 4.5: Final Subnet Allocation

Subnet	Address Range	Prefix
Subnet 1	14.24.74.0 – 14.24.74.127	/25
Subnet 2	14.24.74.128 – 14.24.74.191	/26
Subnet 3	14.24.74.192 – 14.24.74.207	/28
Unused	14.24.74.208 – 14.24.74.255	Reserved

Observation

Allocated addresses:

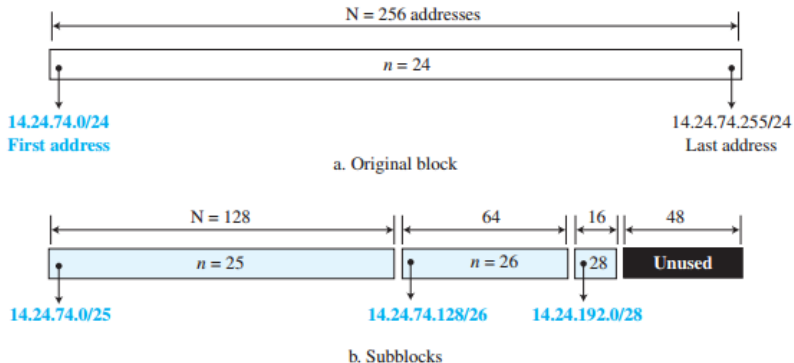
$$128 + 64 + 16 = 208$$

Remaining addresses:

$$256 - 208 = 48$$

These addresses are reserved for future expansion.

Figure 4.36 *Solution to Example 4.5*



Address Aggregation (Route Summarization)

Address Aggregation

- Also called:
 - Route Summarization
 - CIDR Aggregation
- Combines multiple smaller address blocks into one larger block.
- Reduces routing table size.
- Improves routing efficiency.

Working

- ICANN allocates a large block to an ISP.
- ISP divides it into smaller blocks.
- External routers advertise only the larger summarized block.
- ISP internally forwards packets to the correct customer.

Example 4.6: Address Aggregation

Scenario

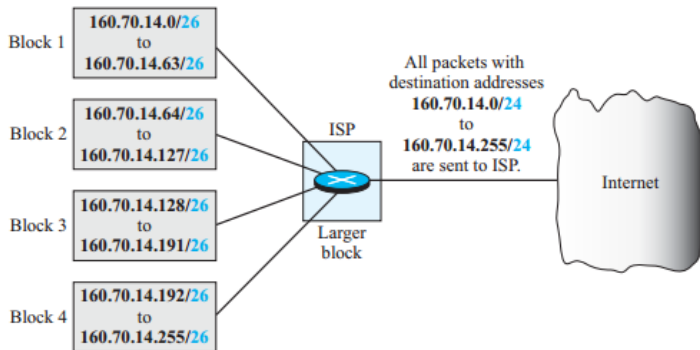
- ISP assigns four small CIDR blocks to four organizations.
- Instead of advertising four separate routes,
- ISP advertises one larger summarized block.

Advantages

- Smaller routing tables.
- Faster routing.
- Reduced routing overhead.
- Better scalability of the Internet.



Figure 4.37 *Example of address aggregation*



Special IPv4 Addresses

Address	Purpose
0.0.0.0/32	This-host address (unknown source address)
255.255.255.255	Limited Broadcast
127.0.0.0/8	Loopback Testing
10.0.0.0/8	Private Network
172.16.0.0/12	Private Network
192.168.0.0/16	Private Network
169.254.0.0/16	Link-local (Private)
224.0.0.0/4	Multicast

Important

Private addresses are not routable on the public Internet and are mainly used with NAT.

Summary of IPv4 Addressing

- IPv4 uses 32-bit logical addresses.
- Address space contains over 4 billion addresses.
- Classful addressing is obsolete due to address depletion.
- CIDR provides efficient variable-length addressing.
- Slash notation specifies the network prefix.
- Address masks determine network boundaries.
- Subnetting divides networks into smaller subnets.
- Address aggregation reduces routing table size.
- Special IPv4 addresses support testing, broadcasting, private communication, and multicasting.

Dynamic Host Configuration Protocol (DHCP)

What is DHCP?

- **Dynamic Host Configuration Protocol (DHCP)** is an application-layer protocol that automatically assigns IP configuration to hosts.
- It follows the **Client–Server** communication model.
- DHCP assists the TCP/IP protocol suite at the network layer by automating address assignment.



Why DHCP?

- Without DHCP, network administrators must manually configure every host.
- DHCP automatically assigns configuration parameters whenever a device joins the network.
- Hence, DHCP is often called a **Plug-and-Play Protocol**.

Key Point

DHCP eliminates manual IP configuration and reduces configuration errors.

Features and Applications of DHCP

DHCP Supports

- Permanent IP address assignment.
- Temporary (leased) IP address assignment.
- Automatic address reuse after lease expiration.

Applications

- Enterprise and campus networks.
- Hotels, airports, and public Wi-Fi.
- Internet Service Providers (ISPs).
- Home and office LANs.

Textbook Example

An ISP owns **1000 IP addresses** but can serve nearly **4000 households**, assuming only about one-fourth of the customers access the Internet simultaneously.

Information Provided by DHCP

DHCP provides more than just an IP address.

A host normally requires:

- 1 IP Address
- 2 Network Prefix (Subnet Mask)
- 3 Default Router (Gateway)
- 4 DNS Server Address

Why are these required?

- IP Address → Identifies the host.
- Prefix/Mask → Identifies the local network.
- Default Router → Enables communication with other networks.
- DNS Server → Resolves domain names into IP addresses.

DHCP Message Format

DHCP follows the Client–Server communication model.

- Client sends a DHCP request.
- Server replies with a DHCP response.

Important Fields

- Transaction ID
- Client IP Address
- Your IP Address
- Server IP Address
- Gateway IP Address
- Client Hardware Address
- Options Field



Figure 4.38 *DHCP message format*

0	8	16	24	31	
Opcode	Htype	HLen	HCount		Fields:
Transaction ID					Opcode: Operation code, request (1) or reply (2)
Time elapsed		Flags			Htype: Hardware type (Ethernet, ...)
Client IP address					HLen: Length of hardware address
Your IP address					HCount: Maximum number of hops the packet can travel
Server IP address					Transaction ID: An integer set by client and repeated by the server
Gateway IP address					Time elapsed: The number of seconds since the client started to boot
Client hardware address					Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used
Server name					Client IP address: Set to 0 if the client does not know it
Boot file name					Your IP address: The client IP address sent by the server
Options					Server IP address: A broadcast IP address if client does not know it
					Gateway IP address: The address of default router
					Server name: A 64-byte domain name of the server
					Boot file name: A 128-byte file name holding extra information
					Options: A 64-byte field with dual purpose described in text

DHCP Options Field

Options Field

- Total size = **64 bytes**.
- Carries additional network configuration information.
- Can also contain vendor-specific information.

Magic Cookie

- Value:

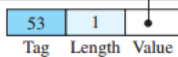
- Used to identify the beginning of the DHCP options.

Each DHCP option contains:

- Tag (1 byte)
- Length (1 byte)
- Value (Variable length)

Figure 4.39 *Option format*

1	DHCPDISCOVER	5	DHCPACK
2	DHCPOFFER	6	DHCPNACK
3	DHCPREQUEST	7	DHCPRELEASE
4	DHCPDECLINE	8	DHCPINFORM



DHCP Operation Overview

When a host joins a network, it does not know:

- Its own IP address
- DHCP server address
- Default gateway
- DNS server address

DHCP assigns these automatically using a **four-step handshake**:

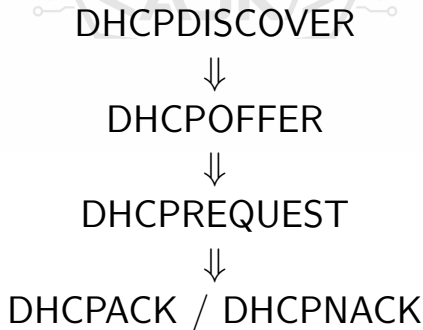
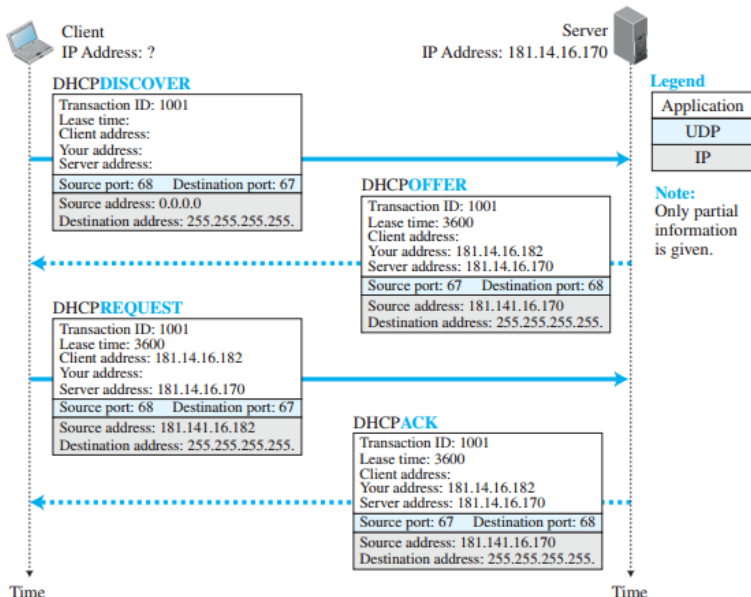


Figure 4.40 *Operation of DHCP*



Step 1: DHCPDISCOVER

When a host starts, it broadcasts a **DHCPDISCOVER** message.

Characteristics

- Transaction ID is initialized with a random value.
- Other fields remain empty because the client knows nothing.
- Encapsulated inside a UDP segment.

UDP Header

- Source Port = 68 (Client)
- Destination Port = 67 (Server)

IP Header

- Source Address = 0.0.0.0 (This Host)
- Destination Address = 255.255.255.255 (Broadcast)

Reason

The client does not know either its own IP address or the DHCP server address.

Step 2: DHCPOFFER

One or more DHCP servers respond with a **DHCPOFFER** message.

Offer Contains

- Offered IP address (Your-IP-Address field)
- DHCP Server IP address
- Lease time
- Other configuration parameters

Packet Details

- UDP Source Port = 67
- UDP Destination Port = 68
- Source IP = DHCP Server Address
- Destination IP = Broadcast Address

Why Broadcast?

Broadcast allows multiple DHCP servers to send offers. The client later chooses the best offer.

Step 3: DHCPREQUEST

The client selects the best offer and sends a **DHCPREQUEST** message.

Purpose

- Accept one server's offer.
- Reject offers from other DHCP servers.

Packet Details

- Source Port = 68
- Destination Port = 67
- Source IP = Newly offered client IP
- Destination IP = Broadcast Address



Reason

Broadcast informs all DHCP servers which offer has been accepted.

Step 4: DHCPACK / DHCPNACK

The selected DHCP server verifies the request.

If successful

- Sends **DHCPACK**.
- Client starts using the assigned IP address.

If unsuccessful

- Sends **DHCPNACK**.
- Client repeats the DHCP discovery process.

Example

If the offered IP address has already been allocated to another client before confirmation, the server sends a DHCPNACK and the client restarts the DHCP process.

DHCP Four-Step Process

DISCOVER → OFFER → REQUEST → ACK

Two Well-Known Ports in DHCP

DHCP uses two well-known UDP ports:

- **Server Port: 67**
- **Client Port: 68**

Why not use an ephemeral client port?

- DHCP server replies using a broadcast message.
- Every host on the local network receives the broadcast.
- If an ephemeral port were used, another application (e.g., DAYTIME client) might accidentally receive the DHCP response.
- Using the dedicated port 68 ensures that only the DHCP client processes the message.

Example

Suppose both a DHCP client and a DAYTIME client accidentally use temporary port 56017.

The DHCP broadcast reaches both applications.

Using port 68 prevents the DAYTIME client from receiving the DHCP message.

DHCP: Multiple Clients and Transaction ID

Multiple DHCP clients may start simultaneously, for example:

- After a power failure
- During network restart
- After system reboot

DHCP distinguishes responses using the **Transaction ID**.

- Generated randomly by the client.
- Included in every DHCP message.
- Allows each client to identify its own reply.

Key Point

Even if several DHCP clients receive broadcast replies simultaneously, the Transaction ID ensures that each client processes only its own messages.

Using FTP and Error Control in DHCP

Using FTP

- DHCPACK does not always contain all configuration information.
- The DHCP server provides the pathname of a configuration file.
- The client retrieves additional information using a File Transfer Protocol (FTP).

Error Control

DHCP uses UDP, which is unreliable.
Therefore DHCP improves reliability by:

- Enabling the UDP checksum.
- Using timers and retransmission.
- Using random timer values to prevent many hosts from retransmitting simultaneously after events like power restoration.

Key Idea

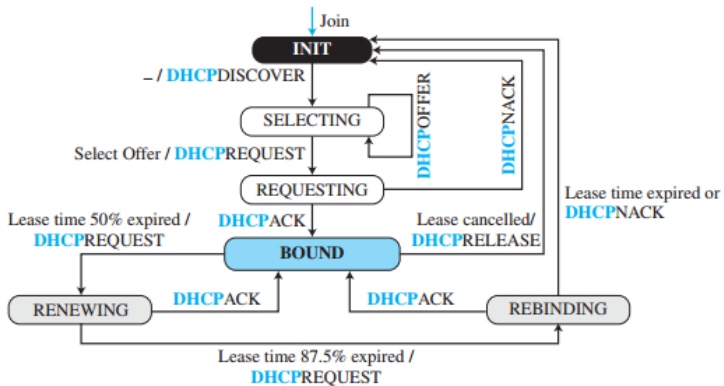
DHCP adds reliability mechanisms even though it operates over UDP.

DHCP Client Transition States

The DHCP client behaves as a finite state machine.

INIT → SELECTING → REQUESTING → BOUND → RENEWING →
REBINDING

Figure 4.41 *FSM for the DHCP client*



DHCP State Description

INIT Client starts and broadcasts DHCPDISCOVER.

SELECTING Client receives one or more DHCPOFFER messages.

REQUESTING Client selects one offer and sends DHCPREQUEST.

BOUND Client receives DHCPACK and begins using the assigned IP address.

RENEWING At 50% of lease time, client requests lease renewal from the server.

REBINDING At 75% of lease time, client broadcasts renewal request to any available DHCP server.

Lease Timers

- Renewal Timer = 50% of lease time
- Rebinding Timer = 75% of lease time
- Expiration Timer = 100% of lease time

DHCP State Transition Summary

State	Purpose
INIT	Client starts DHCP process
SELECTING	Waits for DHCPOFFER messages
REQUESTING	Requests selected IP address
BOUND	Uses assigned IP address
RENEWING	Renews lease with original server
REBINDING	Contacts any DHCP server if renewal fails

Key Takeaways

- DHCP automates IP configuration.
- Uses the DISCOVER-OFFER-REQUEST-ACK sequence.
- Employs UDP ports 67 and 68.
- Uses timers to renew and maintain IP address leases.

Network Address Translation (NAT)

Why NAT is Needed

- ISPs often allocate only a small number of public IP addresses.
- Organizations may have many more computers than available public addresses.
- Most computers do not access the Internet simultaneously.
- Internal communication can use private IP addresses.
- Public addresses are used only for Internet communication.

Example

A company has 20 computers, but at most 4 use the Internet simultaneously.

- Internal communication: 20 private IP addresses.
- Internet communication: Only 4 or 5 public IP addresses.

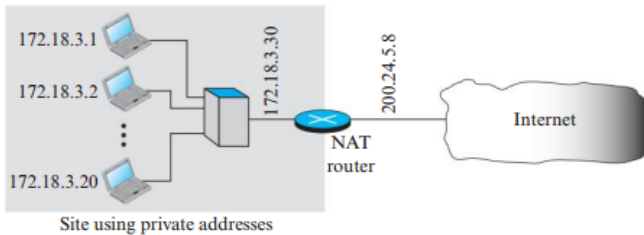
Concept of NAT

- NAT (Network Address Translation) maps private IP addresses to public IP addresses.
- Implemented on a NAT-enabled router.
- Allows many private hosts to share one or more public IP addresses.
- Supports communication between private networks and the Internet.

Key Point

The Internet sees only the public IP address of the NAT router. Private hosts remain hidden.

Figure 4.42 NAT



Address Translation in NAT

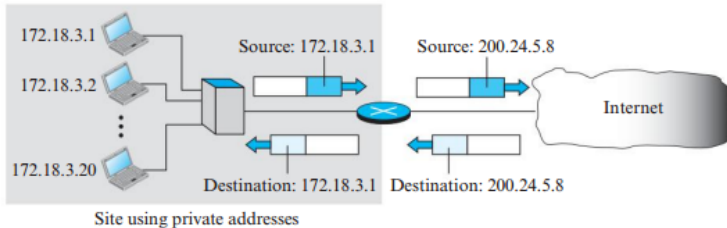
Outgoing Packet

- Private host sends packet.
- NAT replaces the private source IP address with its public IP address.

Incoming Packet

- Packet arrives at NAT router.
- NAT replaces the destination public IP address with the corresponding private IP address.

Figure 4.43 *Address translation*



Translation Table

To correctly forward incoming packets, NAT maintains a **Translation Table**.

The table stores:

- Private IP Address
- External Destination Address

Working

- 1 Outgoing packet is translated.
- 2 NAT records destination information.
- 3 Response packet is matched using the table.
- 4 Original private destination is restored.

NAT Using One Public IP Address

Simplest implementation:

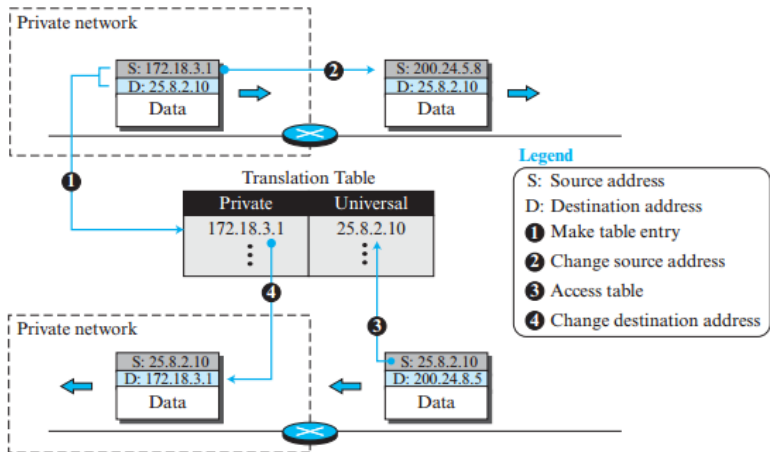
- One public IP address.
- Translation table has two columns:
 - Private Address
 - External Address



Limitation

Only one private host can communicate with a particular external host at a time. Communication must always be initiated from the private network.

Figure 4.44 Translation



Using a Pool of Public IP Addresses

To improve performance, NAT can maintain multiple public IP addresses.

Example

Public Address Pool:

- 200.24.5.8
- 200.24.5.9
- 200.24.5.10
- 200.24.5.11

Advantages:

- Multiple private hosts can access the same external host simultaneously.

Limitations:

- Limited by pool size.
- Number of simultaneous sessions is still restricted.

Using Both IP Addresses and Port Numbers (PAT)

To support many simultaneous connections, NAT also translates port numbers. Translation table contains:

- Private IP Address
- Public IP Address
- Source Port
- Destination Port
- Transport Protocol (TCP/UDP)



Benefit

Different hosts can communicate with the same external server simultaneously because each connection uses a unique port number.

Example of Port Address Translation (PAT)

Two private hosts:

- 172.18.3.1
- 172.18.3.2

Both access the HTTP server:



NAT assigns different source ports:

Private Host	Public Port	Server
172.18.3.1	1400	25.8.3.2:80
172.18.3.2	1401	25.8.3.2:80

Incoming packets are identified using the destination port (1400 or 1401) and forwarded to the correct private host.

Advantages and Limitations of NAT

Advantages

- Conserves public IPv4 addresses.
- Allows many devices to share one public IP.
- Hides internal network structure.
- Supports private addressing.

Limitations

- Breaks true end-to-end communication.
- Some applications require NAT traversal techniques.
- Adds translation overhead at the router.

Exam Point

Most modern home and office routers use NAT with Port Address Translation (PAT).

Summary of NAT

- NAT translates private IP addresses into public IP addresses.
- Enables Internet access using limited public IPv4 addresses.
- Implemented on a NAT-enabled router.
- Uses translation tables for packet forwarding.
- Can operate with:
 - Single public IP
 - Pool of public IPs
 - IP + Port translation (PAT)



Key Takeaways

- Conserves IPv4 address space.
- Provides basic network hiding.
- Widely used in home and enterprise networks.

4.2.3 Forwarding of IP Packets

Forwarding is the process of sending an IP packet toward its destination through the appropriate next hop.

- Internet consists of interconnected networks (links).
- Each router forwards packets to the next hop until the destination is reached.
- Forwarding decisions depend on the forwarding table.

Two forwarding methods in IP

① Forwarding based on Destination Address

- Traditional IP forwarding.
- Uses destination IP address.

② Forwarding based on Label

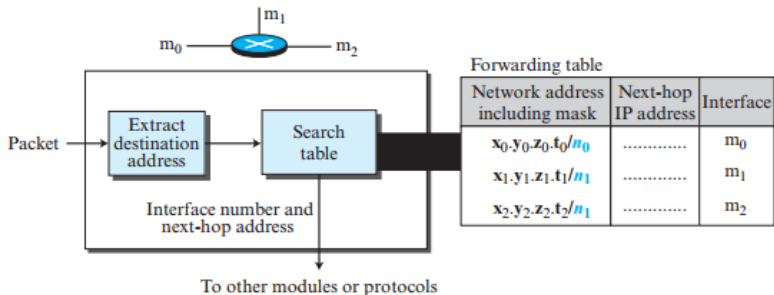
- Used in MPLS.
- Uses packet labels instead of destination addresses.

Forwarding Based on Destination Address

In traditional IPv4 forwarding:

- Every router maintains a **Forwarding Table**.
- When a packet arrives:
 - ① Read destination IP address.
 - ② Search forwarding table.
 - ③ Find matching network.
 - ④ Obtain:
 - Next-hop address
 - Outgoing interface
 - ⑤ Forward packet.

Figure 4.45 *Simplified forwarding module in classless address*

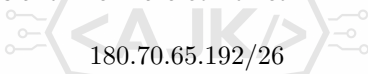


Classless Forwarding Table

A forwarding table in CIDR contains:

- 1 Network Address
- 2 Prefix Length (/n)
- 3 Next-Hop Address
- 4 Interface Number

Usually, Network Address and Prefix are combined:



instead of

$$Network = 180.70.65.192, \quad Mask = /26$$

For every arriving packet:

- Apply the mask.
- Compute the network address.
- Compare with entries.

Forwarding Algorithm

Router searches the forwarding table row by row.

For every entry:

- ❶ Apply prefix mask (/n).
- ❷ Obtain network address.
- ❸ Compare with table entry.
- ❹ If matched:
 - Retrieve next-hop address.
 - Retrieve outgoing interface.
- ❺ Otherwise continue searching.

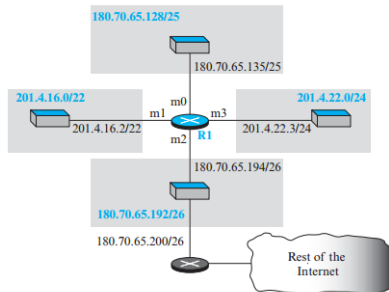
Usually, the last entry is:

Default Route

used when no previous entry matches.

Example 4.7: Forwarding Table for Router R1

Figure 4.46 Configuration for Example 4.7



Network/Mask	Next Hop	Interface
180.70.65.192/26	—	m2
180.70.65.128/25	—	m0
201.4.22.0/24	—	m3
201.4.16.0/22	—	m1
Default	180.70.65.200	m2

Example 4.8: Prefix-Bit Representation

Instead of writing addresses in dotted decimal, routers may store only prefix bits.

Prefix Bits	Interface
10110100 01000110 01000001 11	m2
10110100 01000110 01000001 1	m0
11001001 00000100 00011100	m3
11001001 00000100 000100	m1
Default	m2

Observation

- Longer prefix $\Rightarrow$ Smaller network.
- Shorter prefix $\Rightarrow$ Larger network.

Example 4.9: Packet Forwarding

Destination Address

180.70.65.140

Router operation:

- 1 Apply first mask (/26)

Result:



No match.

- 2 Apply second mask (/25)

Result:

180.70.65.128

Match found.

- 3 Packet forwarded through:
 - Interface = m0

Address Aggregation

In classless addressing, forwarding tables can become very large because every allocated block requires an entry.

Address Aggregation (Route Summarization)

- Combines multiple contiguous address blocks into one larger block.
- Reduces the number of forwarding table entries.
- Speeds up packet forwarding.
- Improves routing scalability.

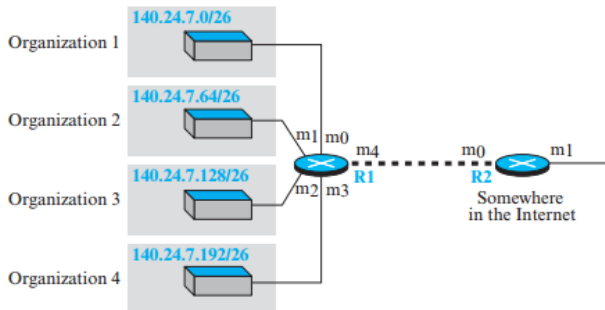
Example

Router R1 maintains separate entries for four organizations.
Router R2 stores only one aggregated entry:

140.24.7.0/24

Any packet destined for this range is forwarded to R1, which performs detailed routing.

Figure 4.47 Address aggregation



Forwarding table for R1

Network address/mask	Next-hop address	Interface
140.24.7.0/26	-----	m0
140.24.7.64/26	-----	m1
140.24.7.128/26	-----	m2
140.24.7.192/26	-----	m3
0.0.0.0/0	address of R2	m4

Forwarding table for R2

Network address/mask	Next-hop address	Interface
140.24.7.0/24	-----	m0
0.0.0.0/0	default router	m1

Longest Prefix (Mask) Matching

Sometimes a smaller subnet is geographically located elsewhere.
To route correctly, routers use the **Longest Prefix Matching Rule**.

- Forwarding table is sorted from
 - Longest prefix
 - To shortest prefix
- Router always selects the entry with the longest matching prefix.

Example

Suppose destination address is

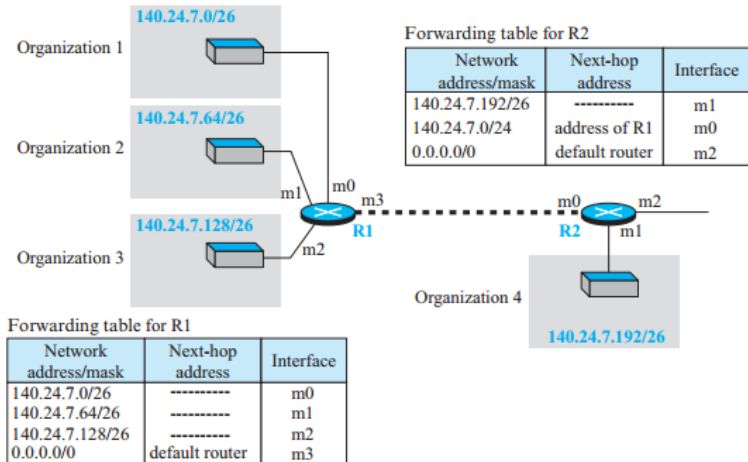
140.24.7.200

Matching entries:

- 140.24.7.192/26
- 140.24.7.0/24

The router chooses

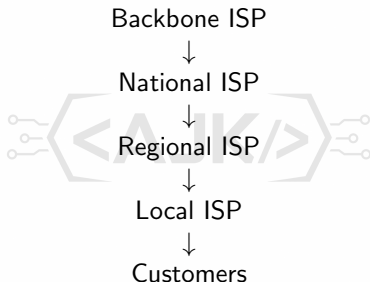
Figure 4.48 Longest mask matching



Hierarchical Routing

Hierarchical routing reduces forwarding table size by organizing networks into multiple levels.

Internet hierarchy:



- Routers outside the hierarchy know only large address blocks.
- Internal routers perform detailed routing.
- Reduces routing complexity.

Example 4.10: Hierarchical Routing

A Regional ISP receives

120.14.64.0/18

which contains

16,384

addresses. The Regional ISP divides the block into four /20 subnets:

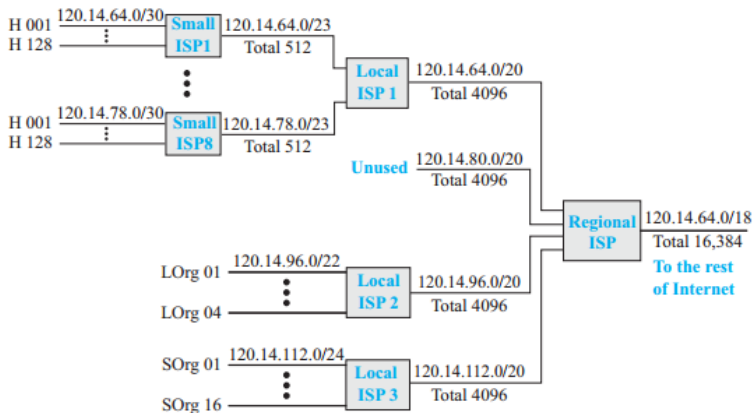
- Local ISP 1
- Reserved block
- Local ISP 2
- Local ISP 3



Further hierarchy:

- Local ISP1 → Small ISPs (/23)
- Small ISP → Households (/30)
- Local ISP2 → Large Organizations (/22)
- Local ISP3 → Small Organizations (/24)

Figure 4.49 Hierarchical routing with ISPs



Benefits of Hierarchical Routing

Example hierarchy:



Advantages:

- Smaller forwarding tables.
- Faster lookup.
- Easier management.
- Better scalability.

Example

Routers outside the Regional ISP need only one entry:

120.14.64.0/18

The Local ISP performs detailed forwarding internally.

Geographical Routing

Forwarding tables can be reduced even further by geographical routing.

The global address space is divided into large geographical regions.

Example:

- America
- Europe
- Asia
- Africa
- Australia



Routers outside a continent require only one entry for that continent.

Benefit

Large numbers of networks are represented using a single forwarding table entry.

Forwarding Table Search Algorithms

Forwarding tables must be searched efficiently.

Longest Prefix Match

- Table divided into buckets according to prefix length.
- Router first searches the longest prefix.
- If not found, searches progressively shorter prefixes.

More efficient data structures include:

- Binary Trees
- Trie (Prefix Tree)
- Other optimized search structures

Observation

Trie-based lookup significantly reduces search time compared to linear table searching.

Forwarding Based on Label

Traditional IP forwarding uses the **destination IP address**. However, modern high-speed networks use **labels** for faster forwarding.

Connectionless Forwarding

- Based on destination IP address.
- Router searches the forwarding table.
- Search operation takes more time.

Connection-Oriented Forwarding

- Based on labels attached to packets.
- Switch directly accesses forwarding table.
- Faster than destination-address lookup.

Routing vs. Switching

- Routing = Searching the forwarding table.
- Switching = Direct table access using labels.

Example 4.11: Longest Prefix Search

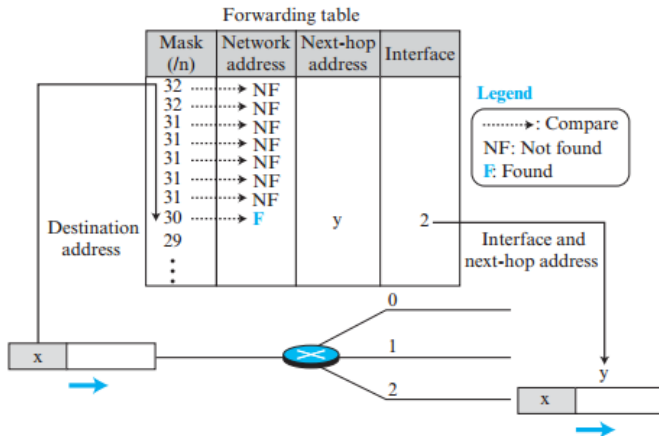
Traditional IP routers perform forwarding using the **Longest Prefix Match** algorithm.

- ➊ Receive destination IP address.
- ➋ Apply mask to compute network address.
- ➌ Compare with forwarding table entries.
- ➍ Find the matching prefix.
- ➎ Extract:
 - Next-hop address
 - Outgoing interface
- ➏ Forward the packet.

Observation

Every forwarding decision requires searching multiple forwarding table entries.

Figure 4.50 Example 4.11: Forwarding based on destination address



Example 4.12: Label Switching

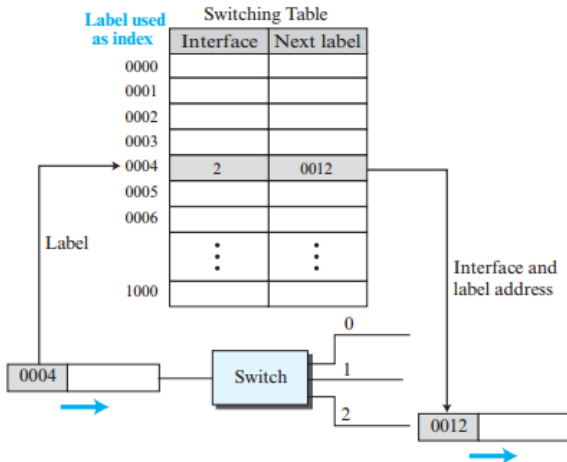
Instead of searching using destination addresses:

- Packet carries a label.
- Label acts as an index into the switching table.
- Required information is obtained immediately.

Key Advantage

Label lookup is significantly faster than destination-address lookup because it performs direct table access instead of searching.

Figure 4.51 Example 4.12: Forwarding based on label



Multi-Protocol Label Switching (MPLS)

During the 1980s, vendors developed routers capable of label switching.

The IETF standardized this technology as

MPLS (Multi-Protocol Label Switching)

MPLS routers can function as both:

- Router (forwarding using destination address)
- Switch (forwarding using labels)

Advantages

- Faster forwarding
- Efficient traffic engineering
- Supports Quality of Service (QoS)
- Enables scalable hierarchical routing

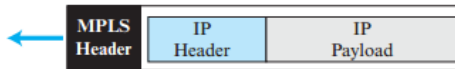
MPLS Packet Encapsulation

IPv4 packets cannot directly carry labels.

Solution:

- Encapsulate the IPv4 packet inside an MPLS packet.
- Add an MPLS header between:
 - Data Link Layer
 - Network Layer
- Entire IPv4 packet becomes the MPLS payload.

Figure 4.52 *MPLS header added to an IP packet*



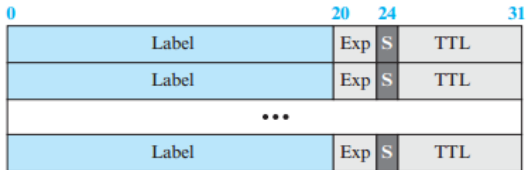
MPLS Header Format

Each MPLS subheader is 32 bits (4 Bytes).

Header fields:

- **Label (20 bits)**
 - Used as index into switching table.
- **Exp (3 bits)**
 - Reserved for experimental/QoS purposes.
- **S (1 bit)**
 - Indicates last label in the stack.
 - $S = 1 \Rightarrow$ Bottom of Stack.
- **TTL (8 bits)**
 - Prevents infinite looping.
 - Decrement at every MPLS router.

Figure 4.53 *MPLS header made of a stack of labels*



Hierarchical Switching in MPLS

MPLS supports multiple labels arranged as a stack.

Each label represents one level of forwarding.

Example:

- Top Label
 - Used for forwarding between organizations or ISPs.
- Bottom Label
 - Used for forwarding within an organization or subnet.

Benefits

- Supports hierarchical routing.
- Faster forwarding.
- Better scalability.
- Efficient backbone routing.

Summary: Forwarding of IP Packets

- IP forwarding can be based on:
 - Destination Address
 - Label (MPLS)
- CIDR forwarding uses longest-prefix matching.
- Address aggregation reduces forwarding table size.
- Hierarchical and geographical routing improve scalability.
- MPLS introduces label-based forwarding for high-speed packet switching.
- MPLS routers combine routing and switching capabilities.

Exam Points

- Longest Prefix Match
- Address Aggregation
- Hierarchical Routing
- MPLS Header Fields
- Difference between Routing and Switching

4.2.4 Internet Control Message Protocol (ICMPv4)

Problem with IPv4

- IPv4 provides no mechanism for:
 - Error reporting
 - Error correction
 - Host or router status queries

Examples:

- Router cannot find a route.
- TTL becomes zero.
- Datagram fragments are missing.
- Destination host becomes unreachable.

Solution: ICMPv4

- Companion protocol to IPv4.
- Operates at the Network Layer.
- Provides:
 - Error reporting

ICMPv4 in TCP/IP

ICMP messages are not sent directly over the data-link layer.

Communication process:

- ① ICMP creates the message.
- ② Message is encapsulated inside an IPv4 datagram.
- ③ IP Protocol field = 1 (indicates ICMP payload).
- ④ Datagram is transmitted normally.

Important

ICMP assists IPv4 but does not replace IPv4.

ICMPv4 Message Categories

ICMP messages are divided into two categories.

1. Error Reporting

- Destination Unreachable
- Source Quench
- Redirect
- Time Exceeded
- Parameter Problem

2. Query Messages

- Echo Request
- Echo Reply
- Timestamp Request
- Timestamp Reply

ICMPv4 Message Format

Each ICMP message contains:

- 8-byte Header
- Variable-size Data section

Common Header Fields:

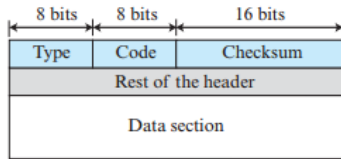
- Type
- Code
- Checksum

Data Section:

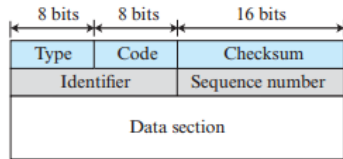
- Error messages:
 - Original IP Header
 - First 8 bytes of original data
- Query messages:
 - Query-specific information



Figure 4.54 General format of ICMP messages



Error-reporting messages



Query messages

Type and code values

Error-reporting messages

- 03:** Destination unreachable (codes 0 to 15)
- 04:** Source quench (only code 0)
- 05:** Redirection (codes 0 to 3)
- 11:** Time exceeded (codes 0 and 1)
- 12:** Parameter problem (codes 0 and 1)

Query messages

- 08 and 00:** Echo request and reply (only code 0)
- 13 and 14:** Timestamp request and reply (only code 0)

Note: See the book website for more explanation about the code values.

ICMP Error Reporting

ICMP only reports errors.

It does NOT

- Correct errors
- Retransmit packets

Error correction is handled by higher-layer protocols (TCP, applications).

Error messages are always sent to:

Original Source Host

because only source and destination IP addresses are available.

Rules for Generating ICMP Errors

ICMP follows strict rules.

It does NOT generate an error message for:

- ❶ Multicast packets
- ❷ Special addresses
 - Loopback
 - This-host address
- ❸ Another ICMP error message
- ❹ Any fragment except the first fragment

Reason

Prevents excessive ICMP traffic and endless error-message loops.

ICMP Error Messages

Message	Type	Purpose
Destination Unreachable	3	Destination cannot be reached
Source Quench	4	Congestion notification
Redirect	5	Better router available
Time Exceeded	11	TTL expired / Fragment timeout
Parameter Problem	12	Invalid IP header/options

Destination Unreachable (Type 3)

Most commonly used ICMP error message.

Uses Codes 0–15.

Examples:

- Host unreachable
- Network unreachable
- Port unreachable
- Protocol unreachable

Example:

- HTTP server is down.
- Router returns:

Destination Host Unreachable

Other ICMP Error Messages

Source Quench (Type 4)

- Indicates network congestion.
- Sender should reduce transmission rate.

Redirect (Type 5)

- Source selected an incorrect router.
- Router forwards packet.
- Router informs source about a better default router.

Time Exceeded (Type 11)

- TTL becomes zero.
- Fragment reassembly timeout.

Parameter Problem (Type 12)

- Invalid IP header.
- Missing or incorrect IP options.

ICMP Query Messages

Used for diagnostics and testing.

They always occur in pairs.

Request	Reply
Echo Request (8)	Echo Reply (0)
Timestamp Request (13)	Timestamp Reply (14)

Applications:

- Test host availability
- Measure Round Trip Time (RTT)
- Check clock synchronization

Echo Request and Echo Reply

Purpose:

- Check whether a host/router is alive.
- Used by the Ping utility.

Communication:

- 1 Sender sends Echo Request (Type 8).
- 2 Receiver replies with Echo Reply (Type 0).

Useful for:

- Connectivity testing
- Measuring delay
- Network troubleshooting

Timestamp Messages

Timestamp Request (Type 13)

- Sender includes current time.

Timestamp Reply (Type 14)

- Returns:
 - Original timestamp
 - Receive time
 - Reply time

Applications:

- Calculate one-way delay
- Calculate round-trip delay
- Synchronize clocks

Example 4.13: Ping Utility

Ping uses:

Echo Request + Echo Reply

Working:

- 1 Send Echo Request.
- 2 Wait for Echo Reply.
- 3 Measure Round Trip Time (RTT).

Information displayed:

- Sequence Number
- TTL
- RTT
- Packet Loss

Sample Output

```
6 packets transmitted 6 received 0rtt min/avg/max = 1.90/1.95/2.04 ms
```

Example 4.14: Traceroute Utility

Traceroute discovers the path from source to destination.

Uses:

- Time Exceeded (Type 11)
- Destination Unreachable (Type 3)

Procedure:

- 1 Send packet with $TTL = 1$.
- 2 Router 1 drops packet.
- 3 ICMP Time Exceeded received.
- 4 Increase TTL.
- 5 Repeat until destination reached.

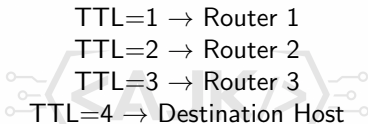
Destination replies with:

Destination Port Unreachable

which indicates tracing is complete.

Traceroute Working

Example:

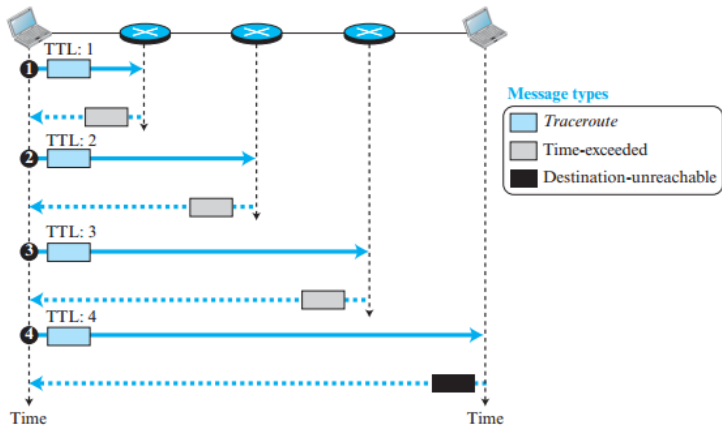


Each router sends an ICMP Time Exceeded message.

The destination sends an ICMP Destination Unreachable message.

Traceroute also calculates RTT for every hop.

Figure 4.55 *Example of traceroute program*



Example 4.14: Sample Traceroute Output

Traceroute Output

```
2 ceneric.net 3.069 ms
3 satire.net 3.071 ms
4 alpha.printers.com 5.922 ms
```

Interpretation:

- 4 devices reached.
- RTT measured at every hop.
- Larger RTT indicates busy routers or longer paths.

Summary: ICMPv4

- Companion protocol to IPv4.
- Provides:
 - Error reporting
 - Network diagnostics
 - Query services
- Error messages:
 - Destination Unreachable
 - Source Quench
 - Redirect
 - Time Exceeded
 - Parameter Problem
- Query messages:
 - Echo Request/Reply
 - Timestamp Request/Reply
- ICMP powers two important utilities:
 - Ping
 - Traceroute